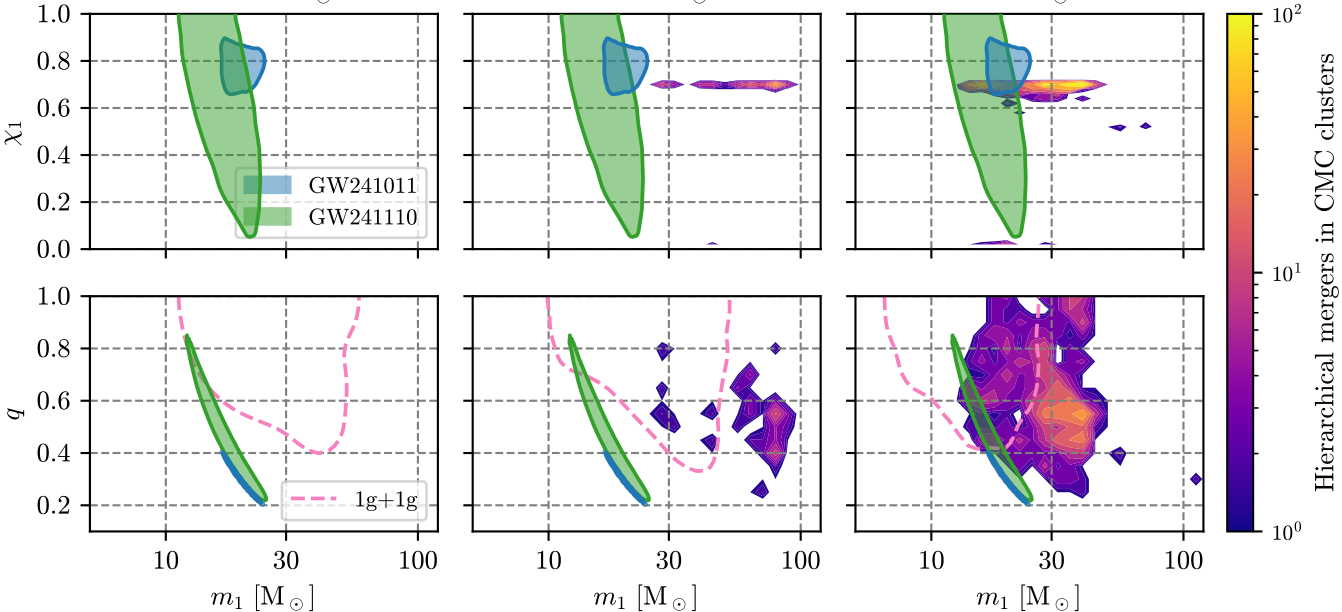

$Z \sim 0.01\ \mathrm{Z}_\odot$
$Z \sim 0.1\ \mathrm{Z}_\odot$
$Z \sim \mathrm{Z}_\odot$
χ_1
q
GW241011
GW241110
1g+1g
$m_1\ [\mathrm{M}_\odot]$
Hierarchical mergers in CMC clusters